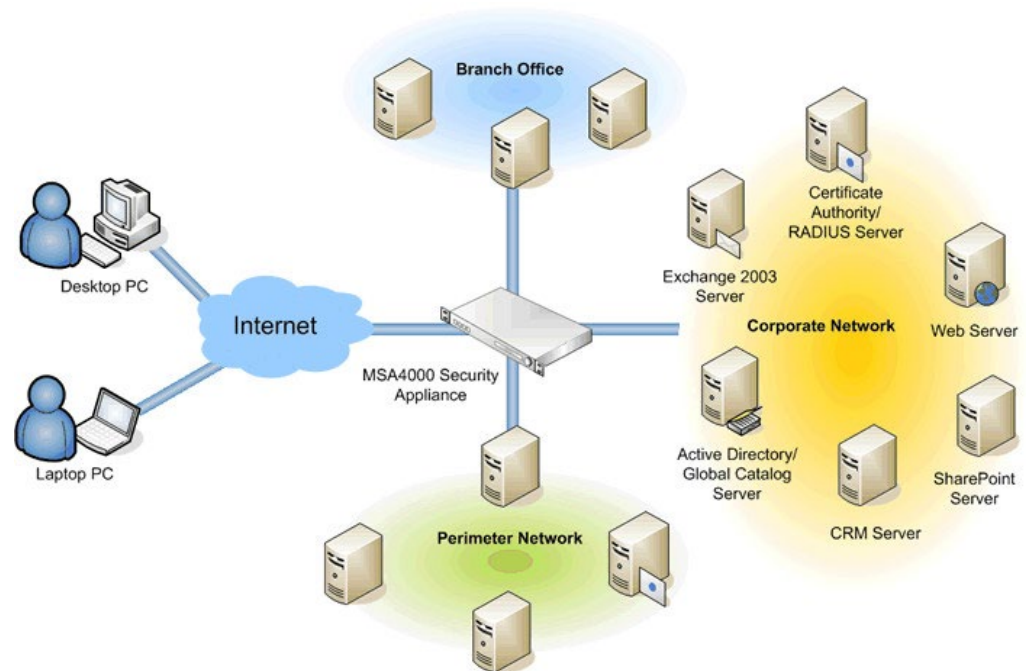
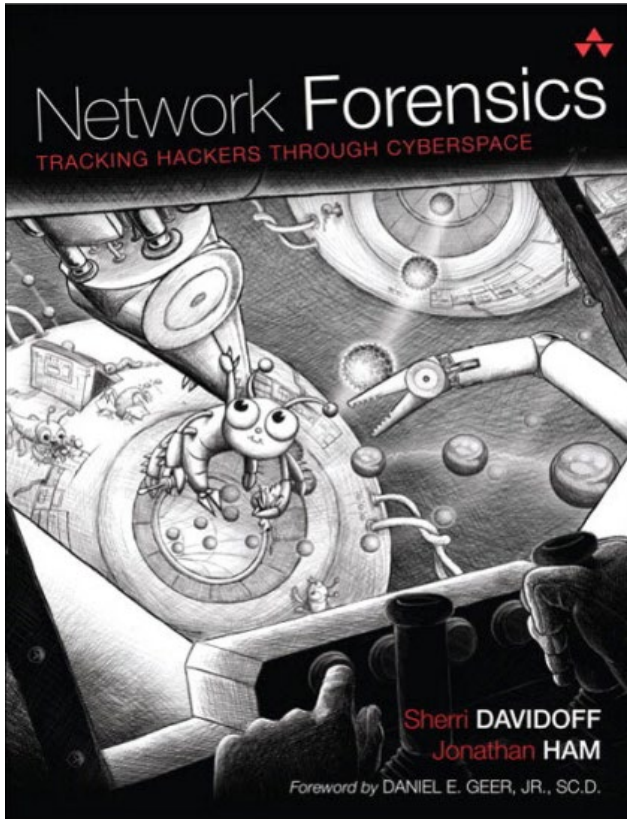


COMP6348: Network Forensics



Lecture #1 – Introduction

Fridays

08:30 – 11:00

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<https://binusmaya.binus.ac.id>

Topics

- ❑ What is Network Forensics
- ❑ Evidence and Data Source
- ❑ Challenge Relating to Network Evidence
- ❑ Investigative Methodology: OSCAR

What is Network Forensics

- ❑ Network forensics is a part of digital forensics with focus to monitor and analyze network traffic.
- ❑ Purposes:
 - ❑ Intrusion Detection/Prevention
 - ❑ Information Gathering
 - ❑ Legal Evidence

Computer vs Network Forensics

- Data is not much change for daily usage
 - Evidence is contained within the file system
 - Easy to perform a forensically sound acquisition
 - Seizing one or several computers would not make deep impact to the business
- Data is much change constantly
 - Evidence sometime exists only in RAM
 - Most network devices does not have non-volatile storage
 - Taking network devices would be problematic

Why We need Network Forensics ?

- As a part of Incident Responses
 - When incident occurred
 - How long incident occurred
 - What data was taken -> any sensitive/confidential information ?
 - How many systems were affected
 - Is the incident still going on ?
- To collect evidences for law enforcement/court
- To find out root cause -> Lesson to learn

What is Evidence

- According to Oxford English Dictionary:
 - Evidence is information or signs indicating whether a belief or proposition is true or valid
 - Information used to establish facts in a legal investigation or admissible as testimony in a law court
-
- Digital Evidence according to National Institute of Justice US:
 - Information stored or transmitted in binary form that may be relied on in court

Evidence and Source

- Computer Forensics
 - HDD, Flashdisk, USB
 - RAM
 - Emails
 - Logs
 - Chat/Messenger
 - Recovered Files
 - Browser History
 - Cache Files
 - CCTV

- Network Forensics
 - Network's devices logs
 - Mirroring Network
 - ISP's Logs
 - IP usage history
 - WiFi usage history
 - CCTV/IP Based Camera

Evidence Categories (1/2)

1. Real

- The murder weapon
- Fingerprint/Footprint
- Signed document (such as paper contract, NDA, etc.)
- Physical HDD/Flashdisk/USB/Computer/Keyboard/Phone

2. Best

- A photo of crime scene
- A copy of signed document
- A file recovered from HDD/Flashdisk/USB/Memory card
- Mirroring network traffic

3. Direct, testimony offered by direct witness of act/acts in question

- “She showed me an inappropriate photos”
- “He told me that he deface that website a few hours ago”

Evidence Categories (2/2)

4. Circumstantial Evidence

- An email signature
- Cache found in defendant's computer/mobile phone

5. Hearsay

- "His parents told me that he received that package before"
- "His friend told me that they did it last night"

6. Business Records

- Employment's contracts/agreement/NDA
- Provider's Signed Agreement
- Invoices and payment records

Challenges Relating to Network Evidence (1/2)

1. Acquisition

- Many possible sources of evidence: wireless access points, web server log, proxy log, server log
- Difficulty gaining evidence, either for political or technical reasons

2. Content

- Data maybe encrypted

3. Storage

- Most network's equipment does not have persistent storage or Non-Volatile storage
- Limited capacity -> limited recorded metadata

4. Privacy

- Personal privacy may be involved -> Unrelated evidence could be exposed

Challenges Relating to Network Evidence (2/2)

5. Seizure

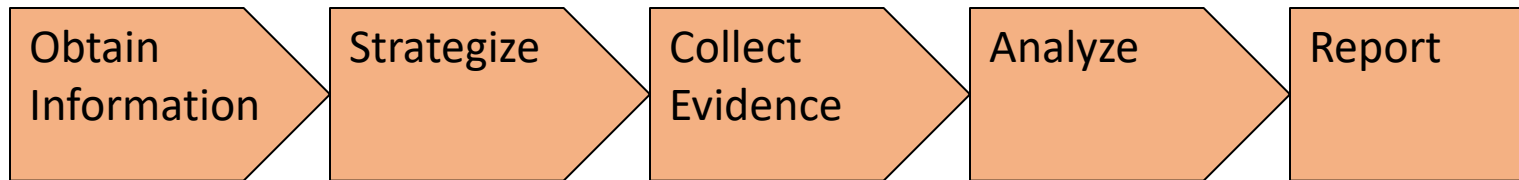
- Seizing equipment in network can be much disruptive
- Investigators must find a way to minimize the impact of network operations

6. Admissibility

- Lawfully Acquired
- Properly Handled (Evidence Custodian)
- Relevant to the case

Investigative Methodology: OSCAR

- Obtain Information
- Strategize
- Collect Evidence
- Analyze
- Report



- Obtain Information
 - Obtain information about the incident and the environment
 - The incident
 - Description of the incident
 - Date, time, and method of incident discovery
 - Persons involved
 - Systems and data involved
 - Actions taken since discovery
 - Legal issues
 - The environment
 - Network topology
 - Available sources of network evidence
 - Organizational structure
 - Resources available (staff, time, funding, equipment)

- Strategize

- Investigators must work efficiently
- List your resources (staff, equipment, time)
- Identify likely sources of evidence and estimate the value and cost to obtain it
- Prioritize your evidence acquisition
- Plan the initial acquisition/analysis

Source of Evidence	Likely Value	Effort	Volatility	Priority
Firewall logs	High	Medium	Low	2
Web proxy cache	High	Low	Medium	1
ARP tables	Low	Low	High	3

- Collect Evidence
 - Document, document, document
 - Lawfully capture evidence
 - Make cryptographically verifiable copies
 - Setup secure storage of collected evidence
 - Establish chain of custody
 - Analyze copies only
 - Use legally obtained, reputable tools
 - Document every step

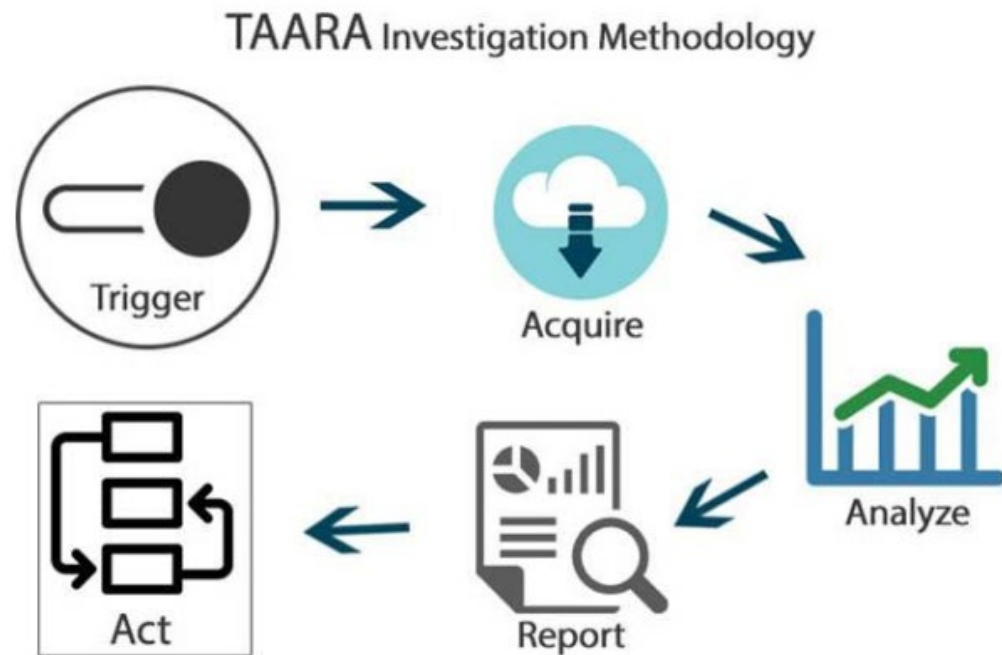
- Analyze
 - Show correlation with multiple sources of evidence
 - Establish a well documented timeline of activities
 - Highlight and further investigate events that are potentially more relevant to incident
 - Corroborate all evidence, which may require more evidence gathering
 - Reevaluate initial plan of attack and make needed adjustments
 - Make educated interpretations of evidence that lead to a thorough investigation, look for all possible explanations
 - Build working theories that can be backed up by the evidence (this is only to ensure a thorough investigation)
 - SEPARATE YOUR INTERPRETATIONS FROM THE FACTS

Investigative Methodology: OSCAR (5/5)

- Report
- Every report must be:
 - Understandable by nontechnical people
 - Complete and meticulous
 - Defensible in every detail
 - Completely factual

Investigative Methodology (TAARA)

- Trigger
- Acquire
- Analysis
- Report
- Action



Source: Learning Network Forensics

Investigative Methodology: TAARA (1/2)

- Trigger
 - The incident that leads to investigation action.
- Acquire
 - Is a part of incident response plan
 - Identifying source of evidences
 - Acquiring evidences
 - Collecting evidences
- Analysis
 - All evidences that already collected would be:
 - Collated
 - Correlated
 - Analyzed

Investigative Methodology: TAARA (1/2)

- Report
 - Based on preceding analysis, a report is produced and present it to stakeholders so the stakeholders can take any necessary action according to recommendations in report.
- Action
 - The action recommended in report is usually implemented in this stage.

- Network forensics is a part of digital forensics which specific to network related. There are some challenges in network forensics such as questioning evidence admissibility and technical issue during acquisition evidence. Therefore, investigators must develop a realistic strategy to meet the investigation goals and available resources.
- With a methodology of investigation in network forensics will help investigators to perform network forensics and achieve investigation goals.